

PREPARED FOR

ACME SURVEYING

LOT 42

T6N, R4W, Mount Diablo Meridian

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
38.3459°N 122.3047°W WGS84	66 ft NAVD88 · MSL	CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)	Zone X 06055C0510FF	13.0° E WMM 2026

APN **038050009000** | ZONING **AP** | ACRES **0.481 ac**

BEST GNSS WINDOW **Tue, Jul 14 · 6:00 AM – 10:00 AM** | PDOP **0.98 · 33 sv** | NEAREST CORS
P261 · 14.1 mi ✓

REPORT CONTENTS

01	7-DAY GNSS FIELD RANKING	202	SITE & SIGNAL REPORT	3
03	BEST STATIC OBS WINDOW	404	EMERGENCY HOSPITALS	5
05	DAY-BY-DAY WEATHER + DRONE	606	AI FIELD ANALYSIS	13
07	WATER LEVELS & GAUGES	1408	FAULTS & SUBSIDENCE	15
09	STRUCTURAL PARAMETERS	1610	SOILS & ENGINEERING	16
11	MINES & MINERALS	1712	NATIVE PLANTS & WILDLIFE	18
13	FCC MICROWAVE PATHS	1914	SOLAR & PASSIVE DESIGN	21
15	AI SITE INTELLIGENCE BRIEF	2316	FIELD MAP · TOPO · DXF SITE PLAN	24



Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 14, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com



This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

14.2 TECU

-4.9 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Tue, Jul 14	0.98	6:00 AM – 10:00 AM	33	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 14, 2026) · Conditions may change – verify before fieldwork



GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window
PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 99°F | Low 58°F | Wind max 12 mph | Gusts 16 mph | Precip 0.00" | Sunrise 5:57 AM | Sunset 8:32 PM

Waning Crescent 0%

GNSS: Avg PDOP **0.98** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.93, 33 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.91		33 sv	IDEAL
7:00 AM	1.00		32 sv	GOOD
8:00 AM	0.88		35 sv	IDEAL
9:00 AM	0.93		33 sv	IDEAL
10:00 AM	1.09		30 sv	GOOD
11:00 AM	1.03		29 sv	GOOD
12:00 PM	0.92		31 sv	IDEAL
1:00 PM	1.20		24 sv	GOOD
2:00 PM	0.97		32 sv	IDEAL
3:00 PM	0.93		33 sv	IDEAL
4:00 PM	0.97		33 sv	IDEAL
5:00 PM	0.98		34 sv	IDEAL
6:00 PM	1.03		28 sv	GOOD
7:00 PM	0.92		31 sv	IDEAL



GO - GOOD CONDITIONS

Partly cloudy | Wind: 12.3 mph (gusts 16.1 mph) | Precip: 0" | Temp: 57.8°–98.7°F



FIELD SAFETY WARNINGS



HIGH HEAT 98.7°F — Monitor crew for heat stress



DRONE WIND ANALYSIS



GO Peak: 12.3/16.1 mph S

* Sunrise: 5:57 AM | Sunset: 8:32 PM | fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	5/6 mph		SE
6 AM	5/7 mph		ESE
7 AM	4/5 mph		ESE
8 AM	1/1 mph		S
9 AM	3/3 mph		S
10 AM	4/4 mph		S
11 AM	6/7 mph		S
12 PM	7/7 mph		S
1 PM	8/9 mph		S
2 PM	10/12 mph		SSW
3 PM	12/16 mph		SSW
4 PM	12/16 mph		SSW
5 PM	9/13 mph		WSW
6 PM	6/11 mph		WSW
7 PM	2/13 mph		SW
8 PM	9/12 mph		SE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

GNSS SURVEY FIELD PLANNING REPORT

JOB: SG-N9D06R | SITE: N/A | LOCATION: 38.3459, -122.3047 (Napa County)

REPORT DATE: Tuesday, July 14, 2026 | SURVEY START: Tuesday, July 14, 2026

VERDICT

GO — Excellent PDOP geometry (0.93–0.98), clear weather, and moderate ionospheric conditions support survey operations on July 14.

BEST SURVEY WINDOW

Tuesday, July 14, 06:00–10:00 AM (local time). PDOP 0.93 with 33 satellites across multi-constellation array. Window aligns with peak geometry and daylight safety. Avoid afternoon heating and potential atmospheric turbulence after 14:00 local time.

BEST 4-HOUR STATIC GNSS SESSION

06:00–10:00 AM, Tuesday, July 14. This 4-hour window delivers optimal satellite geometry (PDOP 0.93) and sufficient daylight. Given moderate ionospheric activity (Kp 4.7, TEC below 30-day median by –4.9 TECU with quality flag caveat), static baseline accuracies should remain within published specifications; however, extend occupation by 10–15% as precaution against ionospheric delay.

DRONE OPERATIONS ASSESSMENT

Approved for visual line-of-sight (VLOS) operations. Wind max 12 mph is well within safe limits for most survey-grade sUAS platforms. Clear skies and low cloud cover optimize RTK positioning performance. No air quality or visibility constraints. Operate between 06:00–18:00 local time to maximize daylight.

FIELD SAFETY WARNINGS

Extreme heat: Peak 99°F. Ensure adequate hydration, sunscreen, and shade breaks. High ambient temperature may impact receiver battery life and personnel stamina. No wildfire threats detected in proximity. Air quality good (AQI 44). Standard safety protocols sufficient.

SITE CONDITIONS & SATELLITE HEALTH NOTES

Land cover is dwarf scrub with low sky obstruction and low multipath risk; recommended elevation mask 13°. GPS PRN 11 currently unhealthy (UNUSUFN); receivers will auto-exclude. PRN 3 and 26 have scheduled maintenance windows outside the survey window. These health advisories do not degrade the favorable PDOP forecast. All-clear on active wildfires within range.

SUMMARY

Conditions favor survey execution on Tuesday, July 14, 2026. Optimal geometry window 06:00–10:00 AM yields PDOP 0.93 with 33-satellite constellation. Moderate ionospheric activity and clear weather present no barriers to static or RTK operations; maintain standard 10–15% occupation time margin for ionospheric margin of safety. Extreme heat is the primary field safety concern. Drone RTK operations are feasible with wind speeds at 12 mph and excellent sky coverage. Proceed with standard pre-survey equipment checks and crew safety briefing.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

Download .txt File

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.82 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	447.42 ft (-19 ft)	466 ft	780K af	80%
Lake Oroville	93.4 mi	859.55 ft (-40 ft)	900 ft	2,836K af	80%
New Melones Lake	100.1 mi	1,019.82 ft (-68 ft)	1,088 ft	1,660K af	69%
Black Butte Lake	101.2 mi	—	228 ft	73K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
-----------	------	-----------	-----------	---------	-------

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.8 ft	695.01 ft	—	100%
Camanche	89 mi	230.2 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	125.3 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	456.8 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Costano Formation (N100CACSTL) is a Miocene-age marine sandstone and siltstone unit that forms one of the primary confined and semi-confined aquifers in Napa County. It was deposited in shallow marine environments and later buried and compacted, creating moderately productive zones. Water availability at this site depends on local structural dip and confining layer integrity; the wide range of water levels observed suggests variable confinement across the mapped well locations.

FORMATION

Name	Napa Valley Costano Formation
Age / Rock	Miocene marine sandstone and siltstone with interbedded shale.
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	100–150 ft (10–15 floors)
Recharge Source	Precipitation on outcrop areas and lateral upland flow from Vaca and Mayacamas Ranges
Transit Time	20–50 years to 150 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 6.8–7.5 (est. USGS/DWR)
Hardness	est. 250–450 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 400–700 mg/L (est. USGS/DWR)

Notes	No sample data — typical for formation. Regional marine-derived salinity expected in deeper, confined zones.
TREND	
Trend	Mixed trend: shallow wells (9.90–21.86 ft BGS, 1977–2022) show minor fluctuation; deeper well (87.71 ft BGS, 2014) indicates thicker vadose zone or local confinement.
Source	Local well registry readings 1977–2022
Seasonal Swing	est. 5–20 ft (est. USGS/DWR based on Napa Valley seasonal recharge pattern)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 **AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU**

The North Coast Ranges in Sonoma County (aquifer N100CACSTL) comprise consolidated bedrock and well-drained alluvial deposits with limited clay sequences, resulting in minimal subsidence potential. Groundwater extraction in this area has remained modest relative to recharge rates, avoiding the intensive pumping cycles that trigger aquifer-system compaction in California's Central Valley and Tulare Lake Basin. For surveyors in this region, historical benchmarks and monumentation are expected to remain stable, allowing conventional survey methods without subsidence corrections.

SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain with limited fine-grained aquitards
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 **AI CLIMATE SUMMARY — RESIDENTIAL DESIGN**

This marine climate demands a well-sealed, moderately insulated envelope that can handle the 92 mph wind loads—focus on continuous air barriers and impact-resistant glazing near exposed facades. Orient primary glazing south and east to capture mild winter sun for passive heating, but minimize west-facing glass to reduce afternoon heat gain and glare during the warmer months. The key passive move for this zone is strategic thermal mass (concrete or masonry) paired with operable windows that allow cross-ventilation during the frequent mild periods, letting you avoid mechanical cooling much of the year while maintaining comfort.

🕒 **AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU**

This coastal marine site presents well-drained gravelly soils suitable for conventional spread footings, though seismic parameters are currently null and may warrant clarification with the local building official. Moderate wind exposure and light snow loads are anticipated, suggesting standard structural design approaches pending confirmation of seismic design category and site-specific geotechnical conditions.

FOUNDATION	
Type	Conventional spread footing anticipated
Basis	Well-drained Coombs gravelly loam with good bearing characteristics suggested for planning. Seismic design category is null — consult local building official to confirm applicable SDC and geotechnical engineer to verify bearing capacity and settlement.
STRUCTURAL IMPLICATIONS	
Seismic	Seismic design category null — classification and corresponding detailing requirements should be confirmed with building official and structural engineer before design advancement.
Wind Exposure	Exposure C (open terrain) anticipated per ASCE 7-22 Risk Category II
Wind	92 mph 3-second gust wind speed anticipated — standard wind load design likely adequate. No hurricane-force exposure anticipated for this coastal marine location.
Snow	Light snow load of 5.8 psf anticipated — standard roof design and drainage likely adequate for this IECC Zone 3C marine climate.

INVESTIGATIONS & INSPECTIONS

Geotech Required	Yes — geotechnical investigation recommended to confirm site class, bearing capacity, settlement potential, and to validate frost depth estimate of 12 inches.
Liquefaction Screen	Seismic parameters null — liquefaction screening should be discussed with geotechnical engineer once SDC is confirmed. Well-drained soils suggest lower liquefaction susceptibility, but subsurface confirmation recommended.
Special Inspections	Special inspections may be required depending on confirmed seismic design category — consult structural engineer and building official.
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning purposes. Geotechnical investigation recommended to confirm actual site class, subsurface stratigraphy, groundwater depth, and soil properties for design-phase analysis.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes	SERIES: Coombs
DRAINAGE CLASS: Well drained	HYDRO GROUP: C
SOIL ORDER: Alfisols	SUBORDER: Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

The Coombs series consists of well-drained gravelly loam soils that formed in alluvial and colluvial materials derived from mixed sources on gently rolling terrain in the interior California valleys. Walking onto this site, you'll encounter a moderately coarse surface layer with abundant gravel interspersed with a clay loam substratum; the site drains readily but the clay-rich lower layers will restrict percolation and can trap moisture seasonally. Developers and surveyors should anticipate moderate excavation resistance due to gravel content and pay attention to fill placement specifications, as the substratum's shrink-swell potential and moisture sensitivity can affect foundation performance.

Alluvial and colluvial parent material, gently rolling foothills and valley margins, typical of oak woodland and grassland regions.

DRAINAGE

Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 2–5% slopes
Ponding Risk	None

ENGINEERING

Bearing Capacity	3,000–4,000 psf (shallow foundations); verify on-site)
Rating	Fair
Shrink-Swell	Moderate — clay loam substratum; minor differential movement possible
Frost Action	Low
Cut Suitability	Good (gravel-rich surface material)
Fill Suitability	Fair (clay substratum — moisture condition and compaction control required)

SEPTIC

Suitability	Conditional
Notes	Perc test required — clay loam substratum may limit infiltration rate and nitrification depth; seasonal perched water possible

HAZARDS

Excavation	Moderate — 20–35% gravel content will slow machine work
Expansive Clay	Low to moderate — clay content present in substratum but not dominant
Erosion	Low on stable 2–5% slopes; moderate gully risk if drainage is disrupted or grade is steepened
Other	None

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdnataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 14, 2026, 5:29 AM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Site Planning Review — Oil and Gas Wells **Location:** 38.3459°N, 122.3047°W (Sonoma County area, California) Well proximity is not a planning concern for this site. The eight wells in the search radius are all plugged/abandoned dry holes, with the nearest at 7.2 miles away—well beyond any actionable threshold for foundation risk, surface easements, or operational impact. No active producers, injection wells, or idle wells exist within relevant distances, and no SB 1137 setback considerations apply to the project.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active producers operate within 5 miles of the site: Basalt Rock Co (2 mi, clay operations), Mt George Pumice Deposit (3.1 mi, pumice), and Silva Pumice Quarry (5 mi, pumice). Expect heavy truck traffic on access roads during operating hours, with potential dust generation from aggregate handling and noise from crushing/processing operations. No inactive deposits within 1 mile warrant Phase 1 environmental assessment.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Assessment **Location:** 38.3459, -122.3047 (Solano County area) ESA Section 7 consultation will be triggered if your project involves federal permits, funding, or actions (e.g., Army Corps wetland permits, FEMA funding, USDA loans), requiring formal biological assessment and incidental take authorization. The most planning-relevant species for this site are **California tiger salamander, vernal pool fairy shrimp, and California red-legged frog**, which indicate the presence of seasonal wetlands and grasslands requiring avoidance or mitigation. A Phase I biological survey should be conducted immediately to map vernal pools, seasonal wetlands, and suitable upland habitat on and adjacent to the project site, with spring field verification to detect listed species presence. If federal involvement is anticipated, early coordination with USFWS and NOAA Fisheries (for Delta smelt) is critical to determine permitting requirements and avoid project delays. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

58	5052	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
common yarrow	<i>Achillea millefolium</i>	2026-05-18
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a **mixed oak woodland transitioning to chaparral**, dominated by blue oak and California black oak with California bay and California buckeye in riparian or mesic microsites. The prevalence of moisture-loving species like Braun's Giant Horsetail, California wild rose, and canyon liveforever alongside drought-tolerant chaparral plants (Phacelia, Dudleya, Pellaea) indicates **variable soil moisture across the property**, likely featuring seasonal seeps or drainage swales adjacent to drier upland slopes. Grading activities should anticipate seasonal saturation in lower-lying areas and plan drainage to prevent erosion in oak woodland zones, as soil disturbance threatens the fine-rooted herbaceous layer and mycorrhizal associations essential to oak health. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

636	155	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		636 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		155 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		1100 obs	
Datura wrightii		57 obs	

SPECIES	OBSERVATIONS
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Northern California Survey Area **Highest-Priority Hazards:** Poison oak (*Toxicodendron diversilobum*) is your dominant concern with 1,439 observations in this region—expect it year-round in brushy and wooded areas. Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*) are the second major threat with 636 observations; they’re active March through November, particularly in dry grass and rocky terrain. Black widow spiders and various wasp species present secondary but serious risks, especially during warm months. **Practical Avoidance:** Wear long pants, closed-toe boots, and long sleeves; use thick gloves when clearing vegetation. Stay on established routes, watch hand placement when climbing or moving brush, and never reach into dark crevices or dense vegetation. Learn to recognize poison oak by its three-leaflet clusters and avoid contact; wash clothing and skin immediately after fieldwork. **Seasonal Considerations:** Rattlesnake activity peaks May–September when temperatures warm; stay alert during morning surveys before snakes move to shade. Poison oak remains hazardous year-round but causes more skin reactions during active growth (spring/summer). Wasps and spiders are most aggressive July–September. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3		
Beam Paths	Closest Beam	Fixed P2P		
TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11			
Total Airways	Jet Routes	Victor Airways	Other Routes			
AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8

Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200

Total Observations

89

Total Species

64

Birds

6

Mammals

13

Reptiles

6

Amphibians

BIRDS (64 species)

COMMON NAME	Scientific Name	OBS
California Towhee	Melozone crissalis	7
California Quail	Callipepla californica	6
Wild Turkey	Meleagris gallopavo	6
Snowy Egret	Egretta thula	6
Western Bluebird	Sialia mexicana	5
Ash-throated Flycatcher	Myiarchus cinerascens	4
Great Egret	Ardea alba	4
Red-tailed Hawk	Buteo jamaicensis	4
Black Phoebe	Sayornis nigricans	4
Northern Flicker	Colaptes auratus	3
Cooper's Hawk	Astur cooperii	3
Western Screech-Owl	Megascops kennicottii	3
Turkey Vulture	Cathartes aura	3
Mallard	Anas platyrhynchos	3
Bushtit	Psaltiriparus minimus	3

+ 49 additional species

MAMMALS (6 species)

COMMON NAME	Scientific Name	OBS
Coyote	Canis latrans	3
Black-tailed Jackrabbit	Lepus californicus	2
Mule Deer	Odocoileus hemionus	2
Domestic Cat	Felis catus	1
Eastern Fox Squirrel	Sciurus niger	1
Striped Skunk	Mephitis mephitis	1

REPTILES (13 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	Sceloporus occidentalis	11
Pacific Gopher Snake	Pituophis catenifer catenifer	7
Northern Pacific Rattlesnake	Crotalus oreganus oreganus	6
California King Snake	Lampropeltis californiae	3
Pond Slider	Trachemys scripta	2

COMMON NAME	Scientific Name	OBS
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	4
Western Toad	<i>Anaxyrus boreas</i>	3
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Rough-skinned Newt	<i>Taricha granulosa</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Characterization: 38.3459, -122.3047 This site supports a diverse wildlife community characteristic of California's foothills or transitional oak-grassland habitat with proximate riparian or wetland resources, as evidenced by the co-occurrence of upland species (California Towhee, California Quail, Western Fence Lizard) and water-associated taxa (egrets, pond turtles, chorus frogs, newts). Several species present warrant regulatory consideration: Foothill Yellow-legged Frog and California Giant Salamander are California Species of Special Concern requiring aquatic habitat protection; Northwestern Pond Turtle and Rough-skinned Newt indicate sensitivity to wetland degradation; and Northern Pacific Rattlesnake and Pacific Gopher Snake suggest intact microhabitat structure. The species assemblage indicates moderate habitat quality with functioning riparian/aquatic corridors, though the presence of introduced species (Domestic Cat, Eastern Fox Squirrel, American Bullfrog, Pond Slider) and common generalists suggests some landscape disturbance and fragmentation typical of rural or exurban settings. Development activities should prioritize retention of native oak woodland, grassland, and riparian habitat connectivity, with particular attention to seasonal wetland and aquatic breeding habitat for amphibians and reptiles of concern. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

* PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↪ excellent
7/12	30.3°	98.8%	↪ excellent
8/12	33.7°	99.3%	↪ excellent
9/12	36.9°	99.8%	↪ annual optimal
10/12	39.8°	99.8%	↪ excellent
12/12	45°	99%	↪ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔍 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Story You're in a great spot for solar—your location gets about 6.6 hours of useful sun per day on average, which is solid, though it dips to just 3.2 hours in winter and jumps to 9 in summer; think of those peak sun hours as your "solar paycheck," and the more consistent your paycheck, the faster panels pay for themselves. Your roof's angle and color are actually bigger players than most people realize: at your latitude, a 38-degree tilt (roughly a 9-in-12 pitch) lines up perfectly with the sun's path, and if your roof is dark, it's absorbing about 301 BTU of heat per hour per square foot in summer—that's like leaving a space heater running all day—so switching to a light color cuts that nearly in half and directly reduces your cooling costs. Watch out for west-facing windows, because afternoon sun pouring through them dumps 104 BTU per square foot in summer, making that side of your home a heat trap; closing blinds in the afternoon is free and powerful. Here's something elegant: a big deciduous tree on the south side of your home gives you 4 free hours of shade in summer (cooling season) while dropping its leaves in winter to let the sun warm your home for 5+ hours during cold months—it's like nature's programmable thermostat. Thermal mass—think of it as the home's ability to soak up heat during the day and slowly release it at night—matters more in homes with heavier construction; your frame-style home holds heat for about an hour, while a more solid structure might hold it 7 hours, which smooths out temperature swings. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing 38.34586°N, 122.30466°W | Napa County, California

Structural Design Drivers & Climate Envelope

This site falls within IECC Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, which immediately signals a mild coastal climate with negligible frost action and no snow load requirement. Wind data is unavailable, and seismic parameters have not been provided in your dataset, making it impossible to identify the controlling lateral load case at this stage. Before finalizing the structural system, seismic design category must be established through USGS mapping or equivalent analysis—Napa County's location on the North American Plate margin near active subduction and transform zones typically drives seismic as the primary lateral design driver in this region, but confirmation is essential. The zero frost depth mitigates frost heave risk entirely, simplifying footing design and eliminating seasonal depth requirements tied to ground freezing.

Geotechnical Foundation Strategy

The site is underlain by Coombs gravelly loam with 2 to 5 percent slopes, classified as Hydrologic Group C with well-drained characteristics. Group C soils typically have slow infiltration rates and moderate permeability; well-drained status indicates internal lateral drainage is adequate and surface ponding risk is low on these slopes. Soil texture data (marked unavailable) must be obtained through Phase I geotechnical investigation to refine bearing capacity estimates and confirm suitability for shallow foundations. The absence of frost depth requirements and well-drained status support conventional spread footing design without subsurface thermal considerations, though standard bearing capacity, settlement, and liquefaction analyses remain mandatory given the depositional environment.

Hazard Assessment

No Quaternary faults are recorded within the search radius, and no active subsidence basin data exists for this location—eliminating differential settlement and fault rupture as primary design drivers. However, this negative finding does not preclude seismic hazard; local fault data must be cross-checked against the most current Alquist-Priolo Earthquake Fault Zone maps and USGS Quaternary fault databases. Wildfire risk data were not provided; given Napa County's documented wildfire exposure, this should be evaluated independently through Cal Fire hazard mapping.

Mining & Site Context

Three active mining operations are located within 5 miles: Basalt Rock Co (clay extraction, 2 miles), Mt George Pumice Deposit (2.8 miles), and Silva Pumice Quarry (5 miles). These operations are not immediately adjacent but warrant confirmation that no subsurface mineral rights or future extraction activity could affect the site. Eight oil and gas wells exist within 10 miles; the nearest (Mobil Oil, 7.2 miles) is plugged and abandoned. No active producers are present, eliminating induced seismicity and subsurface pressure changes as near-term concerns, though well locations should be mapped to confirm no subsurface infrastructure conflicts.

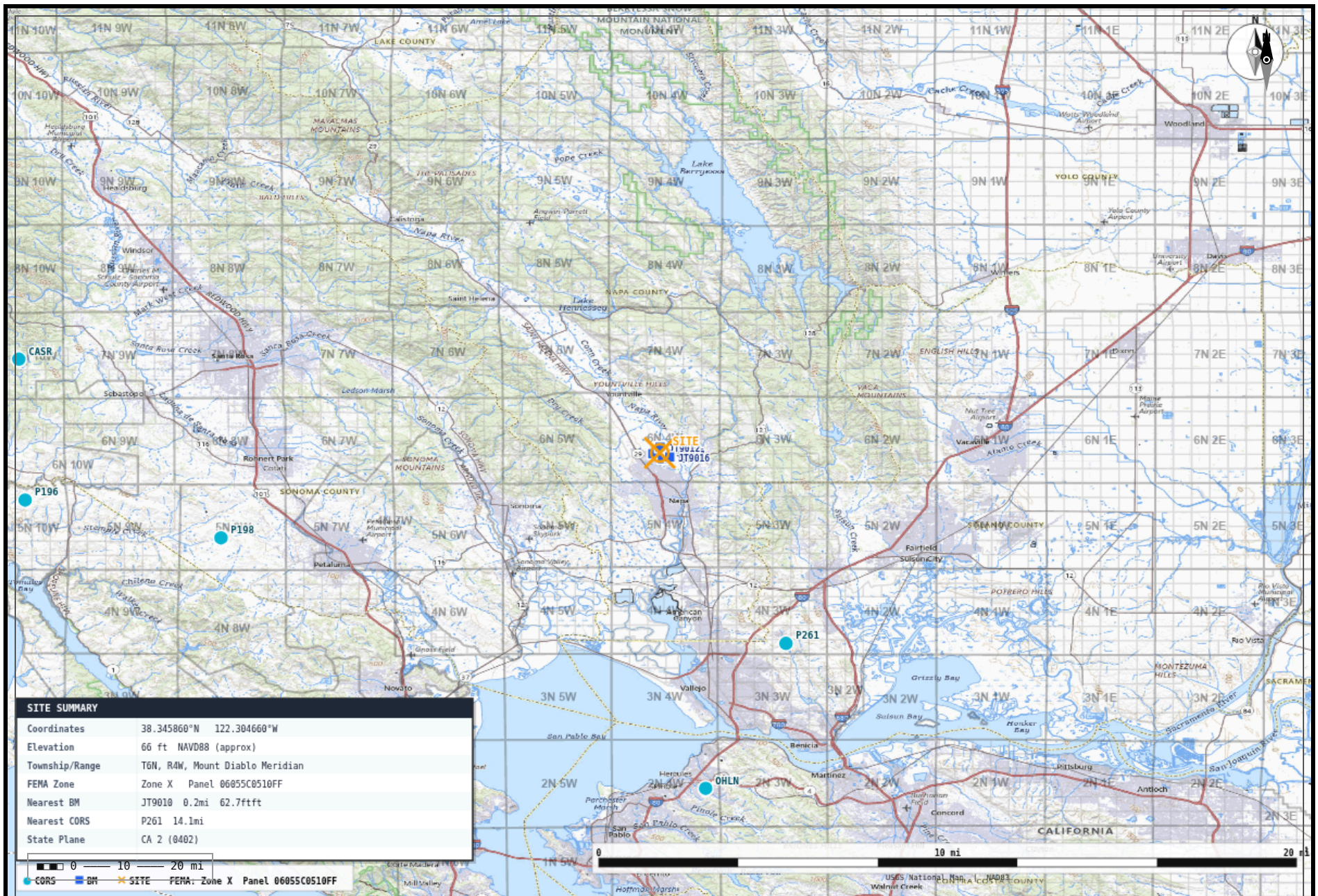
Recommendation

Initiate Phase I geotechnical investigation immediately to confirm bearing capacity, obtain complete soil textiles, and establish seismic design category through USGS hazard zonation and local fault mapping—this will lock structural system assumptions before schematic design.



Copy to clipboard

Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

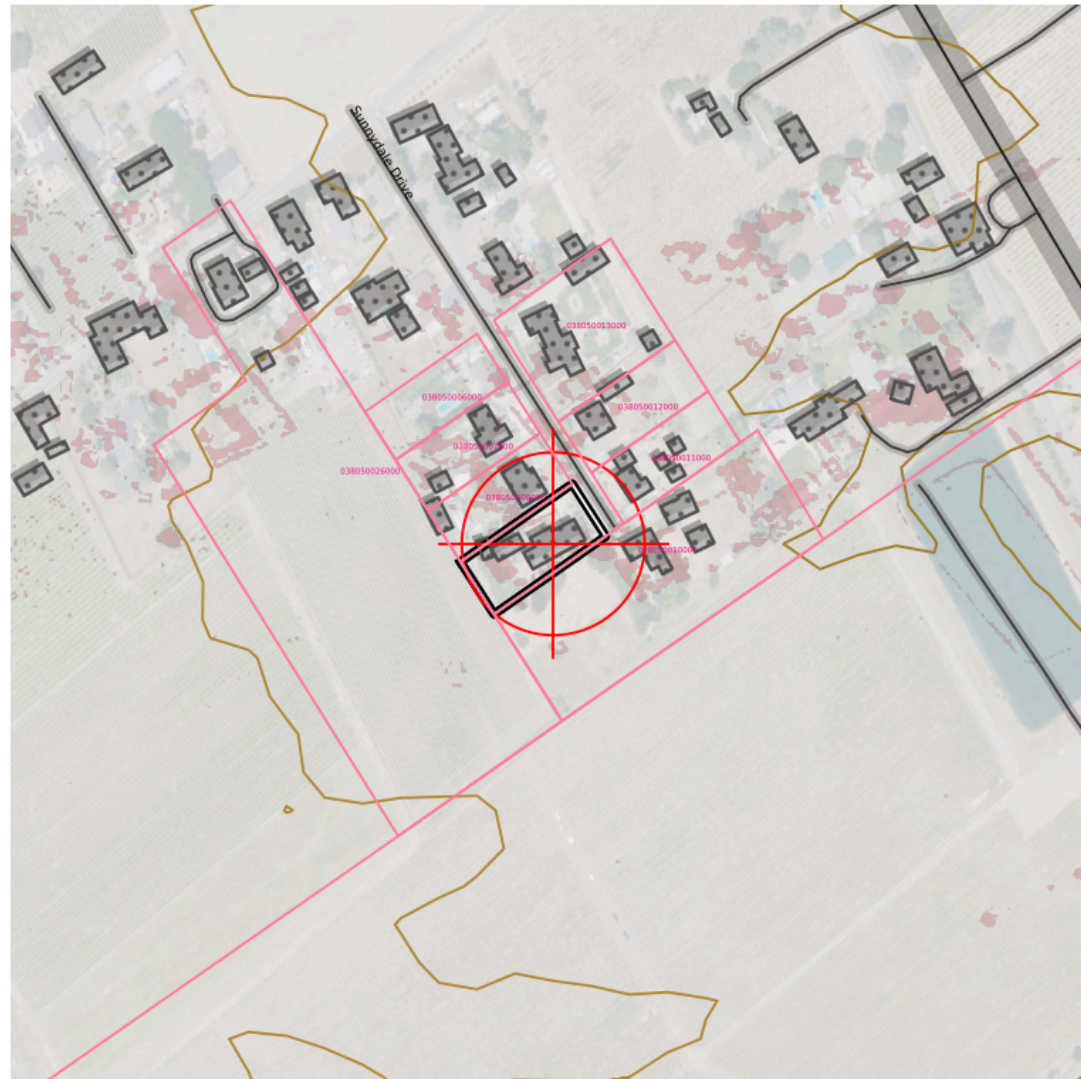


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 14, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polyline (all geometry)



SITE AERIAL – 600 ft × 600 ft

38.34586°N, 122.30466°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

38.34586°N, 122.30466°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

[GEOID18](https://geodesy.noaa.gov/GEOID/GEOID18) and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

